

Supplementary Information

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General Information

All reagents were purchased from commercial suppliers (Acros Organics, Alfa Aesar, Tokyo Chemical Industry, and Sigma-Aldrich) and used without further purification unless otherwise noted. Selected compounds were purified by flash column chromatography on silica gel 60 (0.040–0.063 mm) prior to use.

Analytical thin-layer chromatography (TLC) was performed on silica gel 60 F₂₅₄ glass plates (Merck/Supelco, 20 × 20 cm). Visualization was achieved by UV irradiation (254 or 365 nm) or staining (anisaldehyde, ninhydrin, or KMnO₄) followed by heating when appropriate.

Flow reactions were performed using a Vapourtec E-series system (part no. 50-1319) equipped with a Blue V3 pump tube (part no. 50-1301). For the photochemical reaction (Figure 5), the system was fitted with an organometallic reagent kit (part no. 50-1311). Reactor 1 was a UV–150 photochemical reactor equipped with an LED power supply (part no. 50-1453) and irradiated with a 450 nm LED (part no. 50-1448). The effluent then passed through a second coil reactor irradiated with a blue LED strip. Reactor 1 was maintained at 25 °C using an E-Series cooling module (part no. 50-1314), while reactor 2 was maintained at 25 °C using a water bath and fan cooling. O₂ gas was delivered via a Bronkhorst FLEXI-FLOW compact mass flow controller (FF-C00). For the amide coupling reaction (Figure 6), both reactor coils were held at 95 °C using a water bath. Both setups were fitted with a 7 bar cartridge back-pressure regulator.

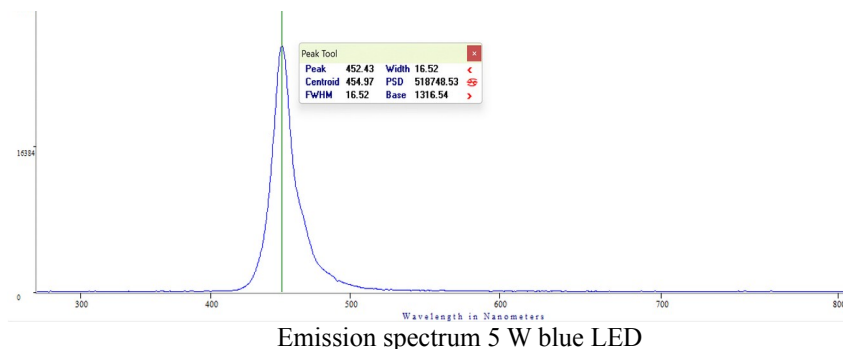
All isolated products were characterized by ¹H and ¹³C NMR spectroscopy and LRMS. Known products were identified by comparison with published NMR data. Copies of all ¹H, ¹³C NMR spectra are provided in the Supporting Information. NMR spectra were recorded at ambient temperature on a JEOL JNM-ECZ500R FT-NMR spectrometer (¹H, 500 MHz; ¹³C{¹H}, 126 MHz). Chemical shifts are reported relative to residual solvent peaks (CDCl₃, δ = 7.26 ppm for ¹H, 77.16 ppm for ¹³C{¹H}; DMSO-d₆, δ = 2.50 ppm for ¹H, 39.52 ppm for ¹³C{¹H}). All ¹³C NMR spectra were broadband ¹H-decoupled. Coupling constants (*J*) are given in Hertz (Hz). Signal multiplicities: br (broad), s (singlet), d (doublet), t (triplet), q (quartet), p (quintet), m (multiplet). Low-resolution mass spectra (LRMS) were recorded on an Advion expression^S CMS spectrometer using electrospray ionization (ESI) in positive-ion mode.

Reaction Setup

Photochemical batch reactions were irradiated using two MR16 5 W blue LED spotlight lamps in the batch reaction setup. Photographs of the lamps and their descriptions are shown below.



MR16 5 W blue LED



Emission spectrum 5 W blue LED

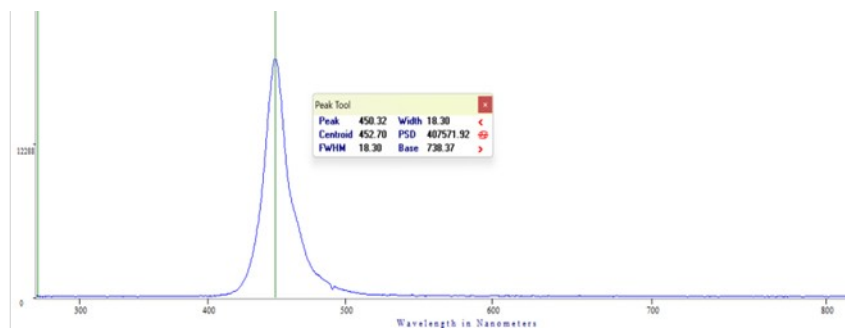
	Specification
Power	5 W
Voltage	12 V
Wavelength	452.4 nm

Figure S1A. Description for 5 W blue LED spotlight lamp.

Photochemical flow reactions were irradiated using Vapourtec 450 nm LED lamp in the standard flow reaction setup. Photographs of the lamps and their descriptions are shown below.



Vapourtec 450 nm LED



Emission spectrum Vapourtec 450 nm LED

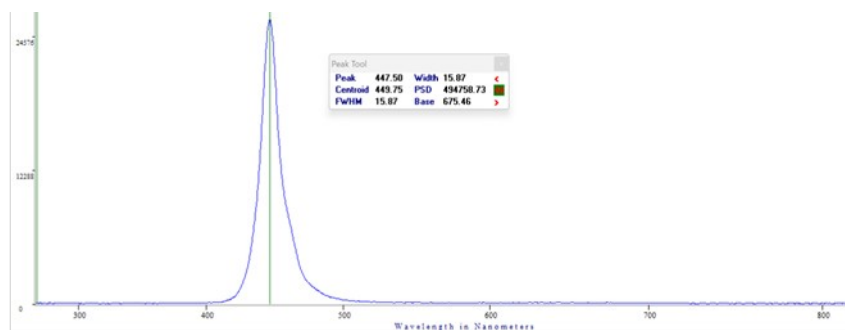
	Specification
Input power	60 W
Radiant power	24 W
Wavelength	450.32 nm

Figure S1B. Description for Vapourtec 450 nm LED lamp.

Photochemical flow reactions were irradiated using blue strip LED in the standard flow reaction setup. Photographs of the lamps and their descriptions are shown below.



Blue strip LED



Emission spectrum blue strip LED

	Specification
Power	10 W
Voltage	24 V
Wavelength	447.50 nm

Figure S1C. Description for blue strip LED.

Experimental Procedures

1. Flow Reactor Setup and Synthesis of Sulfoxide **4a** (Figure 5)

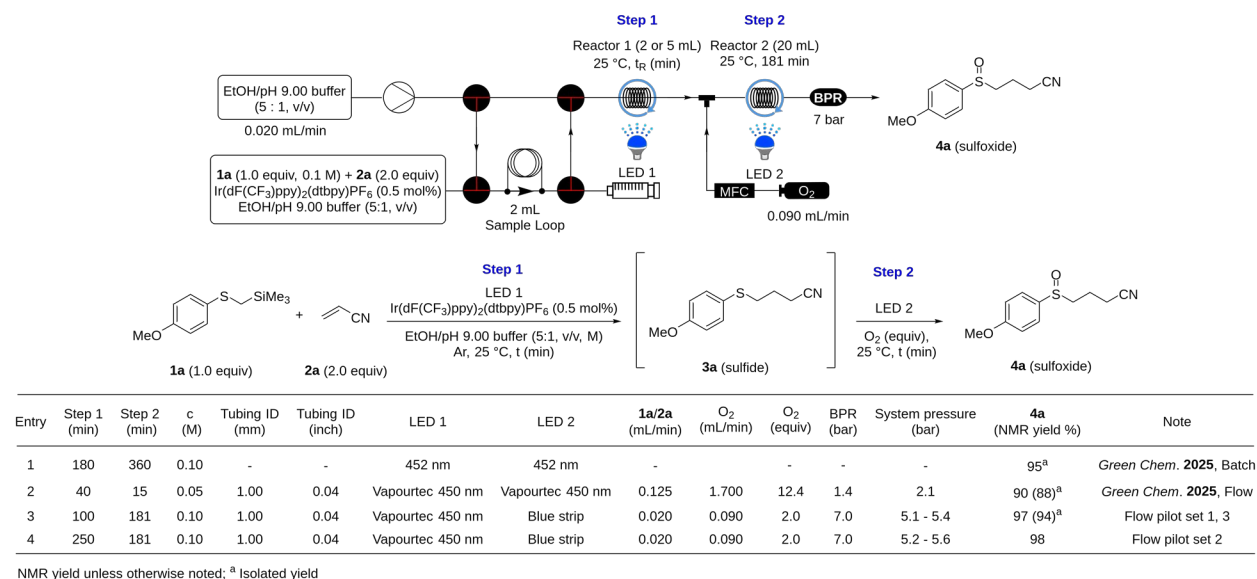


Figure S2. Reactor setup and comparison of batch and flow results for the synthesis of sulfoxide **4a**.

The flow reactor setup used for the experiments in Figure 5 is shown schematically in Figure S2 and photographically in Figure S3.

A Vapourtec E-Series flow system (part no. 50-1319) equipped with a Blue V3 pump tube (part no. 50-1301) was used. Reactor 1 was a UV-150 photochemical reactor fitted with a 2 or 5 mL coil of PFA tubing (0.04 in. i.d.) and irradiated with a 450 nm LED (part no. 50-1448; rated electrical input, 60 W). The reactor temperature was maintained at 25 ± 1 °C. Reactor 2 comprised a 20 mL coil of FEP tubing (0.04 in. i.d.) maintained at 25 ± 1 °C using a water bath and fan cooling and irradiated with a blue LED strip.

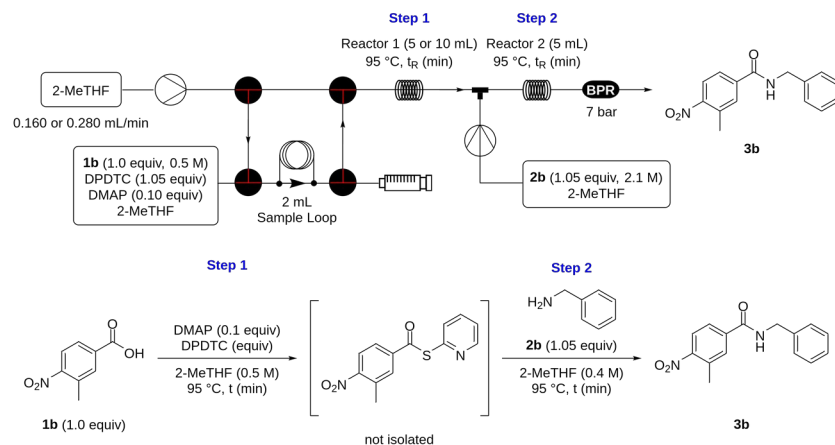
A solution of **1a** (0.2 mmol, 1.0 equiv), **2a** (0.4 mmol, 2.0 equiv), and Ir(dF(CF₃)ppy)₂(dtbpy)PF₆ (0.001 mmol, 0.5 mol%) in EtOH/pH 9.00 buffer (bicarbonate-based buffer, 5:1, v/v; **1a**, 0.10 M) was loaded into a 2.0 mL sample loop and injected into a carrier stream of the same solvent mixture. The liquid flow rate was 0.020 mL min⁻¹. Oxygen was introduced through a T-mixer between the two reactors using a mass-flow controller set to 0.090 mL min⁻¹. A 7 bar cartridge back-pressure regulator was installed downstream of reactor 2, and the recorded system pressure was 5.1–5.6 bar.

The nominal residence times in reactor 1 were 100 and 250 min for the 2 and 5 mL coils, respectively; the nominal residence time reported for reactor 2 was 181 min. An aliquot of the collected reactor effluent was analyzed by ¹H NMR spectroscopy using 1,3-benzodioxole as an internal standard to determine the NMR yield of **4a**. The remaining effluent was concentrated under reduced pressure, and the residue was purified by flash column chromatography on silica gel.



Figure S3. Photograph of the flow reactor setup used for the experiments in Figure 5.

2. Flow Reactor Setup and Synthesis of Amide 3b (Figure 6)



Entry	DPDTC (equiv)	Step 1 (min)	Step 2 (min)	Tubing ID (mm)	Tubing ID (inch)	1b (mL/min)	2b (mL/min)	BPR (bar)	System pressure (bar)	3b (NMR yield %)	Note
1	1.10	30	30	-	-	-	-	-	-	98 (quant) ^a	ACS sustainable Chem. Eng. 2025, Batch
2	1.05	30	30	0.75	0.03	0.043	0.020	4.0	-	97 ^a	ACS sustainable Chem. Eng. 2025, Flow
3	1.05	31.3	25	2.40	0.093	0.160	0.040	7.0	5.2 - 5.3	86 (84) ^a	Flow pilot set 1
4	1.05	35.7	14.3	2.40	0.093	0.280	0.070	7.0	5.5 - 5.8	68	Flow pilot set 2, 3

NMR yield unless otherwise noted; ^a Isolated yield

Figure S4. Reactor setup and comparison of batch and flow results for the synthesis of **3b**.

The flow reactor setup used for the experiments in Figure 6 is shown schematically in Figure S4 and photographically in Figure S5.

A Vapourtec E-Series flow system (part no. 50-1319) equipped with a Blue V3 pump tube (part no. 50-1301) was used with two pump channels. Reactor 1 comprised a 5 or 10 mL coil of ETFE tubing (0.093 in. i.d.), and reactor 2 comprised a 5 mL coil of ETFE tubing (0.093 in. i.d.). Both reactors were maintained at 95 °C in a water bath.

A solution of **1b** (1.0 mmol, 1.0 equiv), DPDTC (1.05 mmol, 1.05 equiv), and DMAP (0.1 mmol, 0.10 equiv) in 2-MeTHF (**1b**, 0.50 M) was loaded into a 2.0 mL sample loop and injected into a carrier stream of 2-MeTHF. The liquid flow rate through reactor 1 was 0.160 or 0.280 mL min⁻¹. At a T-mixer downstream of reactor 1, this stream was combined with a solution of **2b** (1.05 mmol, 1.05 equiv) in 2-MeTHF (**2b**, 2.10 M), delivered through the second pump channel at 0.040 or 0.070 mL min⁻¹, respectively. The combined stream then passed through reactor 2. A 7 bar cartridge back-pressure regulator was installed downstream of reactor 2, and the recorded system pressure was 5.2–5.8 bar.

The nominal residence times in reactor 1 were 31.3 and 35.7 min for the 5 and 10 mL coils, respectively. The nominal residence times in reactor 2 were 25.0 and 14.3 min at combined liquid flow rates of 0.200 and 0.350 mL min⁻¹, respectively. An aliquot of the collected reactor effluent was analyzed by ¹H NMR spectroscopy using 1,3-benzodioxole as an internal standard to determine the NMR yield of **3b**. The remaining effluent was concentrated by rotary evaporation, and the residue was purified by flash column chromatography on silica gel.

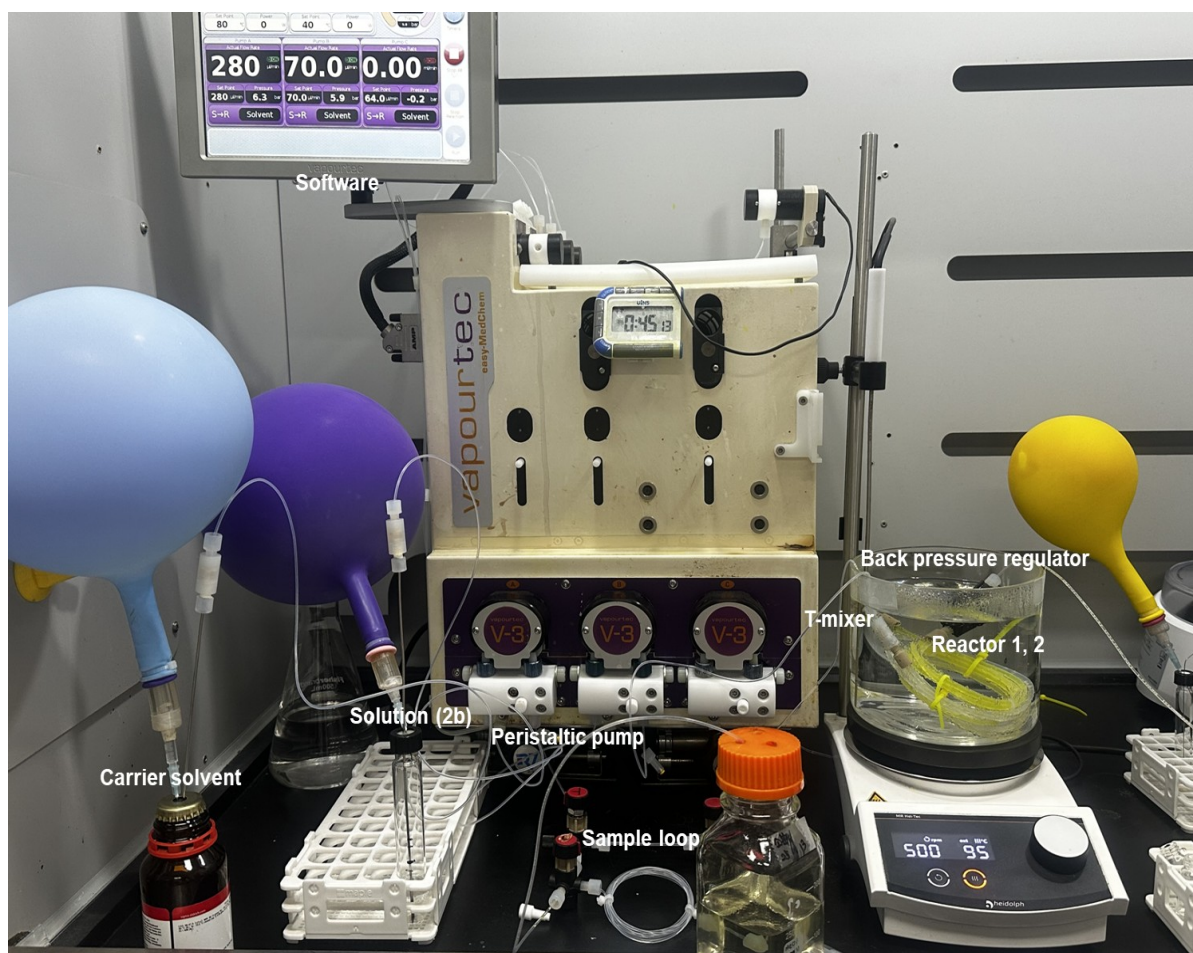
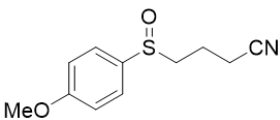
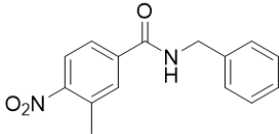


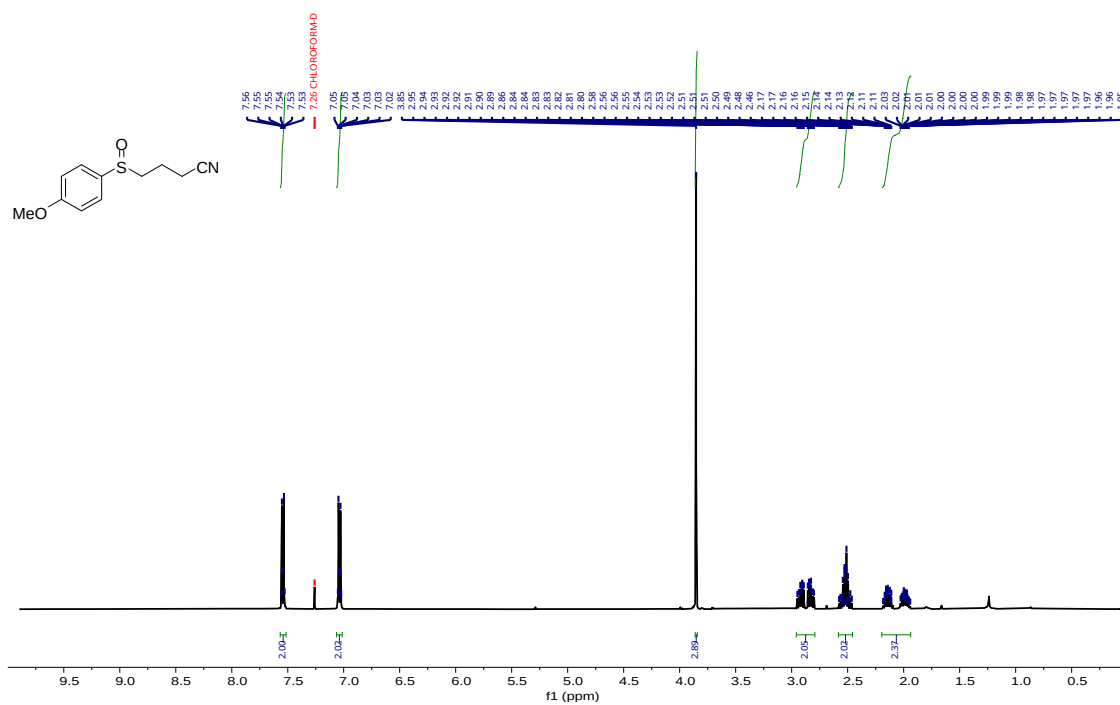
Figure S5. Photograph of the flow reactor setup used for the experiments in Figure 6.

Characterization

 4-((4-methoxyphenyl)sulfinyl)butanenitrile (**4a**) was synthesized according to the standard flow procedure for Figure 5 (ESI section 1) and purified by flash column chromatography on silica gel (n-hexane:EtOAc = 1:4) to afford **4a** as a colourless oil (42.0 mg, 94% isolated yield). R_f = 0.20 (n-hexane:EtOAc = 1:4). ^1H NMR (500 MHz, CDCl_3) δ 7.56–7.53 (m, 2H), 7.05–7.02 (m, 2H), 3.85 (s, 3H), 2.95–2.90 (m, 2H), 2.58–2.46 (m, 2H), 2.19–1.94 (m, 2H); $^{13}\text{C}\{^1\text{H}\}$ NMR (126 MHz, CDCl_3) δ 162.32, 133.66, 125.93, 118.60, 115.11, 55.70, 54.90, 18.67, 16.66; LRMS (ESI) m/z : calcd for $\text{C}_{11}\text{H}_{14}\text{NO}_2\text{S}$ $[\text{M}+\text{H}]^+$: 224.1; found: 224.4; calcd for $\text{C}_{11}\text{H}_{13}\text{NO}_2\text{SK}$ $[\text{M}+\text{K}]^+$: 262.0, found: 261.8.¹

 N-Benzyl-3-methyl-4-nitrobenzamide (**3b**) was synthesized according to the standard flow procedure for Figure 6 (ESI section 1) and purified by flash column chromatography on silica gel (n-hexane:EtOAc = 1:3) to afford **3b** as a pale-yellow solid (227.0 mg, 84% isolated yield). R_f = 0.36 (n-hexane:EtOAc = 1:3). ^1H NMR (500 MHz, CDCl_3) δ 7.95 (d, J = 8.4 Hz, 1H), 7.77 (s, 1H), 7.68 (dd, J = 8.4, 2.0 Hz, 1H), 7.38–7.29 (m, 5H), 6.67–6.60 (br s, 1H), 4.62 (d, J = 5.4 Hz, 2H), 2.60 (br s, 3H); $^{13}\text{C}\{^1\text{H}\}$ NMR (126 MHz, CDCl_3) δ 165.64, 150.89, 138.33, 137.66, 134.16, 131.85, 129.02, 128.08, 128.01, 125.46, 125.03, 44.49, 20.44; LRMS (ESI): calcd for $\text{C}_{15}\text{H}_{15}\text{N}_2\text{O}_3$ $[\text{M}+\text{H}]^+$: 271.1, found: 271.4; calcd for $\text{C}_{15}\text{H}_{14}\text{N}_2\text{O}_3\text{Na}$ $[\text{M}+\text{Na}]^+$: 293.1, found: 293.1.^{2, 3}

(4a) 4-((4-methoxyphenyl)sulfinyl)butanenitrile ¹H NMR (500 MHz, CDCl₃)



References

- [1] J. Park, S. H. Kim, J.-Y. Cho, S. R. Atriardi, J.-Y. Kim, H. Mardhiyah, B. Y. Park, and S. K. Woo, "Batch and flow synthesis of sulfides and sulfoxides using green solvents and oxidant through visible-light photocatalysis, " *Green Chemistry* 27 (2025): 3284.
- [2] J. M. Saunders, E. Ocegüera Nava, J. Li, M. J. Wong, K. M. Freiberg, and B. H. Lipshutz, "Flow-to-Flow Technology: Amide Formation in the Absence of Traditional Coupling Reagents Using DPDTC, " *ACS Sustainable Chemistry & Engineering* 13 (2025): 6646.
- [3] A. Wang, Y. Xie, J. Wang, D. Shi, and H. Yu, "Atom-economic amide synthesis by using an iron-substituted polyoxometalate catalyst, " *Chemical Communications* 58 (2022): 1127.